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ABSTRACT

Hybrid bonding is a pivotal technology for enabling three-dimensional integrated circuits (3D-ICs). Among the foremost challenges facing 3D-IC implementation is thermal management, where a deep understanding of heat conduction across bonded interfaces is essential for addressing heat dissipation and reliability issues. Nevertheless, the thermal conductance of bonded dielectric–dielectric interfaces remains poorly understood. In this study, we employ the low-temperature bonding technique integral to hybrid bonding to fabricate SiO_x–SiO_x interfaces and investigate their thermal boundary conductance (TBC) using time-domain thermoreflectance. Structural characterizations show high-quality bonded interfaces. By fitting the data with an equivalent multilayer thermal model, we establish a lower-limit TBC of 150 MW m^{-2} K^{-1} for the SiO_x–SiO_x interfaces, which corresponds to a thermal resistance lower than that of a 9.2-nm-thick dielectric layer. These findings offer valuable insights into thermal transport in hybrid-bonded structures and provide critical guidance for the thermal design of advanced packaging solutions.

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Three-dimensional integrated circuits (3D-ICs) are promising to extend Moore's law.[1–3] Advanced packaging technologies such as hybrid bonding are critical to achieving high-density interconnects for the heterogeneous integration and interconnection of chips with different manufacturing processes.[4] By high-precision alignment of Cu vias and surrounding dielectric materials (e.g., SiO_x), hybrid bonding achieves ultra-high-density electrical interconnections comparable to monolithic integration performance.[5–8] However, to meet the stringent thermal budget requirements of advanced packaging processes, low-temperature hybrid bonding has become an indispensable technology.[3,9] Although plasma activation facilitates the formation of interfacial covalent bonds to enable bonding at low temperatures, it may also lead to suboptimal interfacial contact, such as van der Waals-like weak interface, at the dielectric–dielectric bonding interface.[10] Such inferior contact can introduce excessively high thermal resistance, which poses a critical challenge to the heat dissipation of the entire packaging system.

As critical signal routes for vertically stacked dies, bonded interconnects which provide high I/O density can also act as heat sources.[11] Meanwhile, as a key component of the heat dissipation path, they generate hotspots, thermal gradients, and thermal strains at specific locations during chip operation, leading to performance degradation of metal pathways and bonding regions.[12–15] Thus, investigating the thermal resistance distribution in these regions is crucial, and the thermal properties of Cu vias and the surrounding dielectrics warrant accurate characterization and optimization. The processing technology and materials differ significantly from those of conventional high-temperature bonding technologies, especially in hybrid bonding processes designed to meet the low thermal budget requirements of chip manufacturing.

Kim *et al.* investigated the thermal transport properties of Cu–Cu bonded interfaces with passivation metals.[16] The thermal properties of Cu–dielectric interfaces have also been studied.[17–19] However, the thermal conductance of the dielectric–dielectric interfaces is much less studied due to the difficulties in thermal measurements, even though these interfaces may exhibit low thermal boundary conductance (TBC) due to voids and van der Waals contacts at the interfaces. The large thermal resistance in hybrid bonded interfaces would degrade the performance of 3D-ICs. Therefore, investigating the thermal properties of the dielectric–dielectric interfaces is not only critical for quantifying thermal resistance distribution at packaging level but also provides key guidance for optimizing low-temperature bonding processes, specifically, on how to avoid weak physical contact and eliminate interfacial voids to achieve robust and strong bonding.

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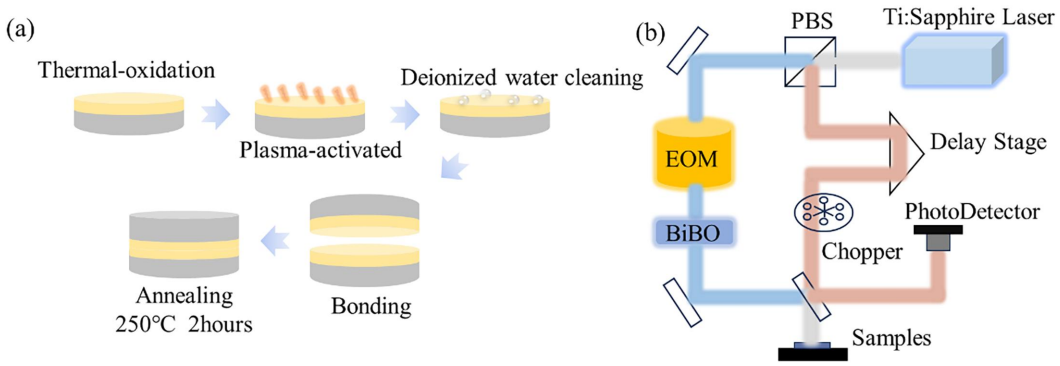


FIG. 1. (a) Schematic diagram of fabrication of SiO_x–SiO_x wafer bonded structure. (b) Schematic diagram of the TDTR measurement.

In this work, we fabricated a SiO_x–SiO_x interface via plasma activation and low-temperature (250 °C) bonding. The SiO_x thin films (≈500 nm) were grown on Si substrates through thermal oxidation in an oxygen environment at 800 °C, ensuring high film quality and uniformity. Figure 1(a) illustrates the detailed fabrication process. Prior to bonding, plasma activation was performed on the sample surfaces. After plasma activation, the samples were rinsed with de-ionized (DI) water for 10 s and subsequently dried with nitrogen gas. Following the bonding process, thermal annealing was conducted at 250 °C for 2 h.

We employed time-domain thermoreflectance (TDTR) technology to characterize the thermal properties of the bonded samples. As shown in Fig. 1(b), a femtosecond pulsed laser (800 nm) is split into a pump beam and a probe beam via a polarized beam splitter (PBS). The pump laser is amplitude-modulated by an electro-optical modulator (EOM) and then frequency-doubled (400 nm) using a bismuth borate (BiB_3O_6, BIBO) crystal, serving as a heat source on the sample surface. In contrast, the probe beam achieves different delay times relative to the pump laser through a mechanical delay stage, which is used to extract the temperature decay after thermal excitation. The thermal penetration depth of TDTR ranges from tens of nanometers to micrometers depending on the thermal diffusivity of the materials and heating frequency, $d_p = \sqrt{\alpha/(\pi f_{\mathrm{mod}})}$, where α is the thermal diffusivity and f_{mod} is the modulation frequency of pump beam. To characterize the SiO_x–SiO_x interface of interest, the SiO_x layer on one side of the interface needs to be thinner than the thermal penetration depth. This ensures that the thermal gradient can reach the interface. Based on the thermal properties of amorphous SiO_x and the modulation frequencies employed in this work, the thermal penetration depth is estimated to range from 165 nm (at 10 MHz) to 280 nm (at 3.37 MHz).

For the bonded structure, the SiO_x layers have a thickness of approximately 500 nm which is thermally thick for TDTR measurements of the interface. Thus, we employed chemical mechanical polishing (CMP) to thin the upper SiO_x layer, as illustrated in Fig. 2. The sample was tilted at a very small angle during CMP to expose all the layers of interest and the SiO_x–SiO_x interface. The samples were cleaned with anhydrous ethanol and de-ionized water to eliminate organic residues generated by CMP on the exposed cross sections. Subsequently, a thin aluminum film (≈80 nm) was deposited on the sample surface as transducer. By focusing the laser spot at different positions and observing the echoes in the in-phase signal curve, the thickness of the upper SiO_x layer was determined by the picosecond acoustic technique.[20] Herein, we selected different thicknesses of the upper SiO_x layer (all around 100 nm) at various modulation frequencies to ensure our measurements penetrate through the bonded interfaces.

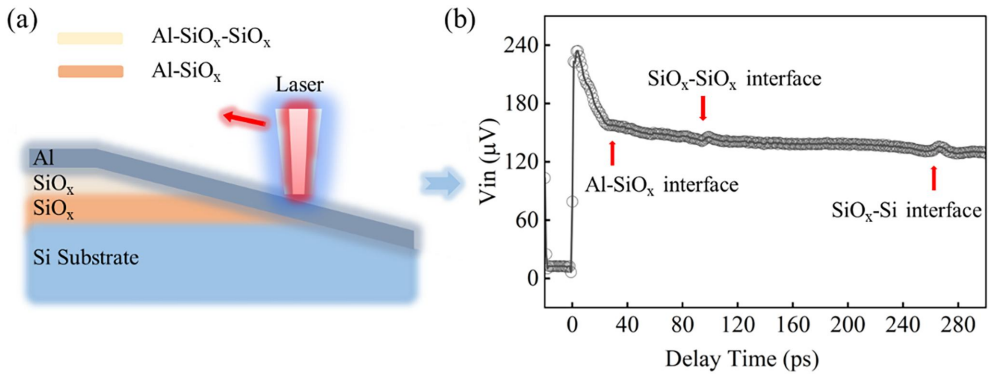


FIG. 2. The sample structure and the corresponding picosecond acoustic measurements. (a) The sample structure after the CMP thinning process and (b) the echoes of the sample in the picosecond acoustic measurements.

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TABLE I. Parameters of TDTR measurements.

Pump/probe radii (μm)	Frequency (MHz)	Thermal penetration depth (nm)	Thickness of upper SiO_x (nm) (with ≈ 8% uncertainty)
15/7.5	10	165	99
15/7.5	5.63	219	96
6.3/6.3	5.02	232	84
15/7.5	3.37	283	136

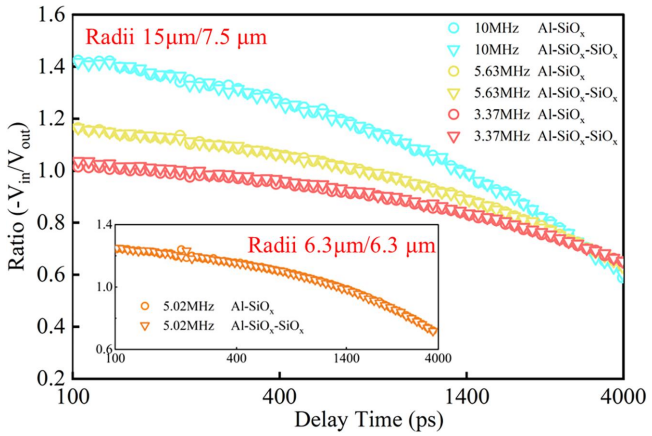


FIG. 3. TDTR ratio data of Al–SiO_x sample structure and Al–SiO_x–SiO_x sample structure with different modulation frequencies and pump/probe radii.

The thermal properties measured by TDTR (including thermal conductivity and TBC) were obtained by fitting the experimentally measured TDTR ratio signal to the analytical solutions of a multilayer thermal conduction model.[21] The thickness of each layer in the multilayer structure needs to be determined in advance. The bottom 500-nm-thick SiO_x layer was thermally thick and thus treated as a bulk substrate in the data fitting. We selected the middle point of the rising peak of the SiO_x–SiO_x signal as the interface position. The thickness of the upper SiO_x layer is the product of the difference of echo positions of the SiO_x–SiO_x interface and the Al–SiO_x interface and the longitudinal sound velocity of SiO_x (≈5800 ms^{-1}), then divided by two.

Two sets of pump/probe laser spot radii were employed in our experiments (15/7.5 μm and 6.3/6.3 μm). At each tested modulation frequency, an appropriate position on the sample was selected according to the echo positions to ensure that the thickness of the upper SiO_x layer was approximately 100 nm and the TDTR heating can penetrate through the bonded interfaces. The ratio curves were then recorded accordingly, with detailed parameters listed in Table I.

For the sample structure adopted in this work, the multilayered configuration involves multiple thermal properties, including the TBC of the Al–SiO_x interface, the TBC of the SiO_x–SiO_x interface (the interface of interest), and the thermal conductivities of the upper and bottom SiO_x layers. Prior to the measurements at the positions listed in Table I, TDTR measurements (10 MHz) were performed on regions of the sample that do not contain the SiO_x–SiO_x interface. These measurements were used to measure the TBC of the Al–SiO_x interface and the thermal conductivity of SiO_x, with both values fitted simultaneously. The fitted TBC of the Al–SiO_x interface was 117 MW m^{-2} K^{-1}, and the thermal conductivity of SiO_x was 1.36 W m^{-1} K^{-1} (with an 8% error), which is consistent with the reported values in the literature.[22] Since the upper and bottom SiO_x layers were fabricated via the same

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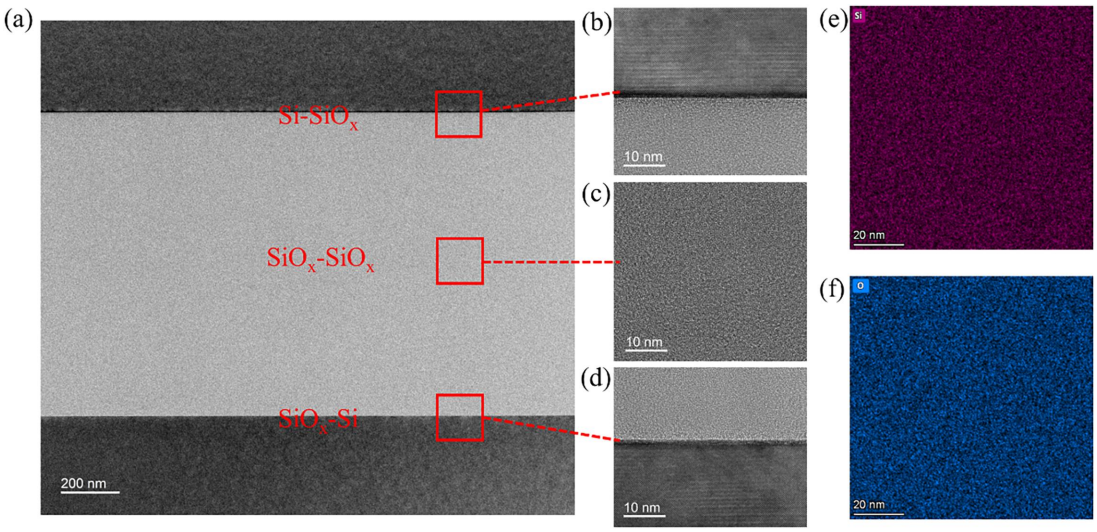


FIG. 4. Transmission electron microscope (TEM) and energy dispersive x-ray spectroscopy (EDS) images. (a) Overall structure image of sample; (b)–(d) Si–SiO_x and SiO_x–SiO_x interfaces; and (e)–(f) silicon (Si) and oxygen (O) elemental distributions.

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process, and the thickness of the upper SiO_x layer at all selected positions in Table I is approximately 100 nm (a thickness free of the size effect for SiO_x), we reasonably assumed that the thermal conductivities of the upper and bottom SiO_x layers are equal and fixed at this fitted value.

Figure 3 shows the results of the TDTR measurements. At the same modulation frequency, two separate measurements were performed: one was conducted on sample regions with the SiO_x–SiO_x interface included in the structure, and the other on regions with only the Al–SiO_x bilayer structure. These comparative measurements were implemented to investigate the effect of the SiO_x–SiO_x interface on the ratio curves, thereby evaluating the effect of the introduced interfacial thermal resistance (ITR, the reciprocal of TBC) at the SiO_x–SiO_x interface. The results demonstrate that there is almost no discernible difference in the measured signals with and without the SiO_x–SiO_x interfacial thermal resistance. This sufficiently indicates that the interface formed via plasma activation and subsequent low-temperature bonding annealing exhibits a considerably low interfacial thermal resistance. It is challenging to extract the TBC of the interface between two low thermal conductivity layers, because the additional interfacial thermal resistance introduced by the SiO_x–SiO_x interface corresponds to an extremely short Kapitza length. Specifically, based on the thermal conductivity of SiO_x (1.38 W m^{-1} K^{-1}), a TBC of 150 MW m^{-2} K^{-1} for this interface corresponds to an equivalent SiO_x thickness of only 9.2 nm. After low-temperature annealing, Si–O–Si covalent bonds form at the SiO_x interface, which significantly improves the properties of this homogenous interface and thus yields a considerably high TBC (the Kapitza length for the SiO_x–SiO_x interface in this work is less than 10 nm), which has a negligible influence on the overall thermal resistance.

To gain further insight into the interfacial properties, high-resolution transmission electron microscopy (HRTEM) and energy-dispersive X-ray spectroscopy (EDS) characterizations were performed on the bonded interfaces.

As shown in Fig. 4(a), the bonded structure comprises both Si–SiO_x and SiO_x–SiO_x interfaces. The SiO_x dielectric layers were prepared via thermal oxidation, a process known to produce high-density amorphous films with good uniformity compared to deposited oxides. For the Si–SiO_x interface, the two constituent materials exhibit distinct crystal structures, which results in a very sharp and well-defined interface in the transmission electron microscope (TEM) images [Figs. 4(b) and 4(d)]. In contrast, the SiO_x–SiO_x bonded interface, formed between two identical amorphous layers, is indistinguishable. Even under HRTEM imaging [Fig. 4(c)], no structural discontinuity can be identified, as shown in Fig. 4(c). After plasma activation, bonding, and low-temperature annealing, covalent bonds (Si–O–Si bonds) are formed at the interfaces.[23] Figures 4(e) and 4(f) show the distributions of the two key elements (Si and O) across the interfacial region. The distribution is uniform, with no discernible differences between the two sides of the interface. This indicates that there is no damage, no severe structural defects (such as voids) at the interface after bonding and annealing. We note that thermal boundary conductance across amorphous/amorphous interfaces is governed by factors beyond structural perfection. In amorphous materials, the absence of long-range periodicity leads to a diffusion-dominated transport mechanism, where the distinction between acoustic and optical phonon modes becomes blurred compared to crystalline systems. Wang *et al.* acknowledge the importance of grain boundaries (GBs) in suppressing thermal transport and have highlighted that even in nanostructured boundaries, the suppression of high-frequency vibrational modes (including optical phonon-like modes) at the boundary can contribute to finite thermal resistance.[24] Therefore, while the structural quality of our bonded interface is high, we conservatively estimate a minimal limit for the thermal boundary conductance, acknowledging that the effective TBC may be higher

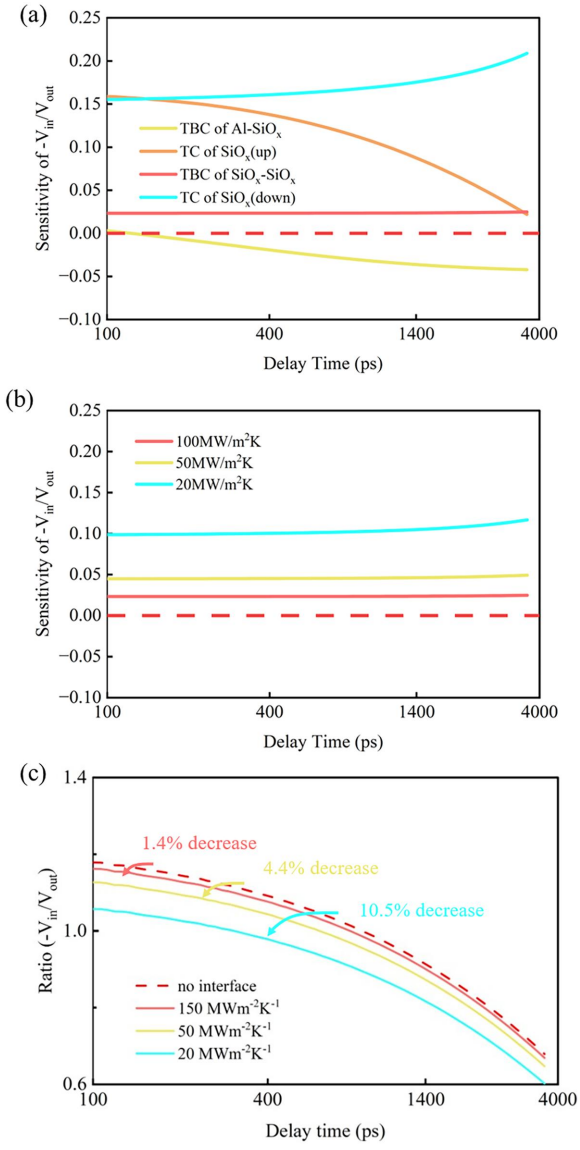


FIG. 5. (a) TDTR sensitivity analysis with the pump/probe radii of 15 and 7.5 μm respectively, and modulation frequency of 5.63 MHz. The thermal conductivities of Al and SiO_x are fixed at 170 and 1.38 W m^{-1} K^{-1}, respectively, and the TBC values of Al–SiO_x and SiO_x–SiO_x interfaces are both set to 100 MW m^{-2} K^{-1}; (b) sensitivity of the SiO_x–SiO_x TBC with values of 100, 50, and 20 MW m^{-2} K^{-1} while keeping other parameters the same. (c) Decrease in the ratio curve relative to the no interface case under different SiO_x–SiO_x TBC values.

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than our current estimate if such mode-specific suppression effects are minimal in our system.

Figure 5(a) presents the calculated sensitivity of different key parameters at the given initial values, where the sensitivity of the SiO_x–SiO_x interface at a TBC of 100 MW m^{-2} K^{-1} is lower than that of all other parameters. As shown in Fig. 5(b), a lower SiO_x–SiO_x TBC corresponds to a higher sensitivity. This is because a lower TBC indicates a larger interfacial thermal resistance introduced at the interface, which imposes a more significant influence and thus results in a higher sensitivity. As shown in Fig. 5(c), when the SiO_x–SiO_x TBC is set to 150, 50, and 20 MW m^{-2} K^{-1}, the ratio curves exhibit decreases of 1.4%, 4.4%, and 10.5%, respectively. We can indicate that the TBC of interest in our experiments is sufficiently high, which is unable to induce measurable differences in the acquired ratio curves. In other words, the ratio curves show almost no discrepancy whether the SiO_x–SiO_x interface is included in the structure or not. The ratio curves in Fig. 3 reflect that the SiO_x–SiO_x interface of interest exhibits a considerably high TBC.

To obtain an equivalent TBC value for the SiO_x–SiO_x interface, the thermal conductivity of SiO_x was fixed at a series of values ranging from 1.25 to 1.5 W m^{-1} K^{-1}, while the TBC values of both the Al–SiO_x and SiO_x–SiO_x interfaces were fitted simultaneously. The corresponding results are presented in Figs. 6(a)–6(d). The thermal conductivity of SiO_x can be considered to lie in the range of 1.25–1.38 W m^{-1} K^{-1}. The equivalent fitting results in Fig. 6 show that the TBC values of the Al–SiO_x interface are all around

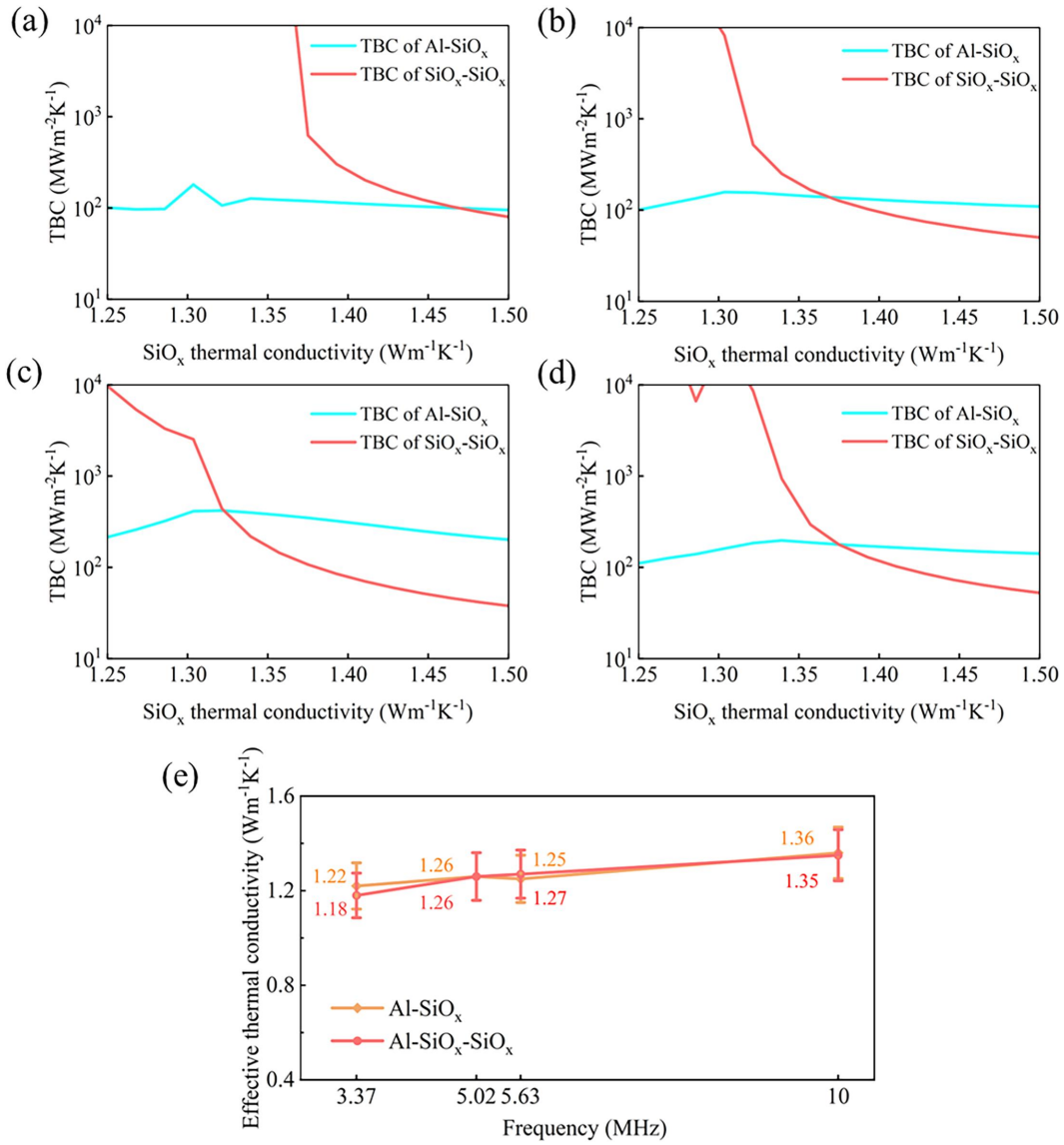


FIG. 6. (a)–(d) Simultaneous fitting of Al–SiO_x and SiO_x–SiO_x interfacial TBC with fixed SiO_x thermal conductivity. (e) Effective thermal conductivity results under assumed two-layer structure. The error bars represent an 8% uncertainty. Al–SiO_x TBC is fixed at 117 MW m^{-2} K^{-1}.

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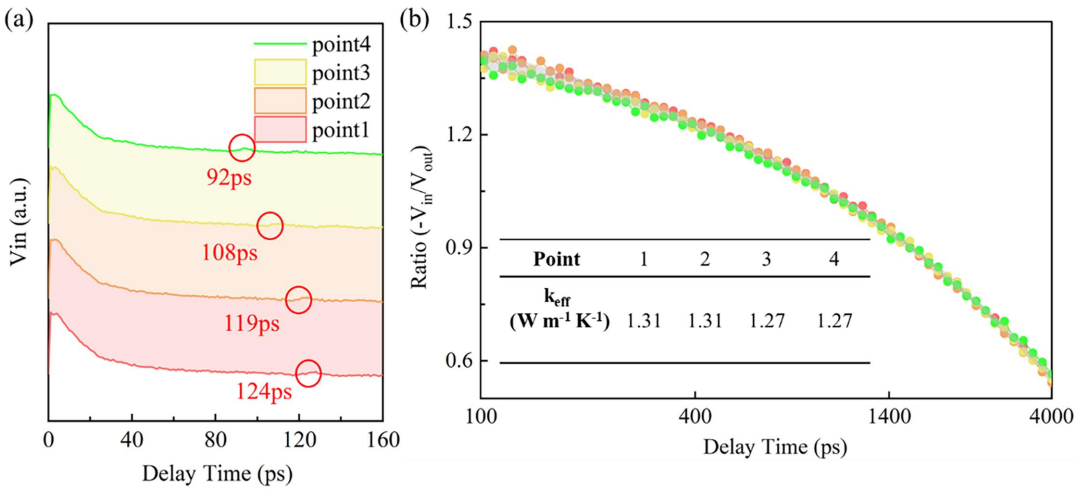


FIG. 7. Measurement results of CVD-deposited SiO_x–SiO_x sample. (a) V_{in} curves at different points and (b) ratio curves and fitted values of effective thermal conductivity.

100 MW m^{-2} K^{-1}, while the TBC values of the SiO_x–SiO_x interface can be regarded as greater than 150 MW m^{-2} K^{-1}.

We fitted the effective thermal conductivity of the structure with the introduced SiO_x–SiO_x interface by simplifying the Al–SiO_x–SiO_x trilayer structure into an Al–SiO_x bilayer structure. As shown in Fig. 6(e), the effective thermal conductivity derived from this bilayer simplification exhibits almost no difference from the fitting results obtained from regions without the SiO_x–SiO_x interface, which is consistent with the nearly indistinguishable ratio curves. All the equivalent fitting results are distributed within an 8% error range relative to 1.25 W m^{-1} K^{-1}, and the slight discrepancies in the results at different modulation frequencies can be attributed to measurement errors.

To evaluate the generality of our findings beyond thermally grown oxides, we fabricated an additional sample using CVD-deposited SiO_x, while keeping the subsequent low-temperature bonding process identical. Interestingly, the CVD SiO_x sample also exhibits a relatively high TBC at the bonded interface. Figure 7(a) shows measurements at different points with varying upper SiO_x thicknesses (points 1–3: Al–SiO_x–SiO_x structure and point 4: Al–SiO_x structure) and Fig. 7(b) displays the ratio curves and fitted values of effective thermal conductivity. The results demonstrate that the CVD SiO_x–SiO_x interface formed through low-temperature bonding similarly exhibits a high TBC that is effectively undetectable within the sensitivity limits of our TDTR measurements.

These findings demonstrate that both thermally grown and CVD-deposited SiO_x layers can form high-quality dielectric–dielectric interfaces with negligible thermal resistance following proper surface activation and low-temperature annealing. The formation of Si–O–Si covalent bonds across the bonded interface ensures structural continuity, resulting in an extremely short Kapitza length that falls below the detection limit of our TDTR measurements. Consequently, the interfacial thermal resistance introduced by the SiO_x–SiO_x interface can be neglected in high-quality hybrid bonding structures. This work extends the applicability of hybrid bonding technology from thermally grown oxides to CVD processes, providing critical insights for thermal management in advanced packaging applications.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Xingqiang Zhang and Liu Chang contributed equally to this paper.

Xingqiang Zhang: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Validation (equal); Visualization (equal); Writing – original draft (lead); Writing – review & editing (equal). **Liu Chang:** Data curation (equal); Formal analysis (equal); Investigation (equal); Writing – review & editing (equal). **Liyi Li:** Conceptualization (equal); Funding acquisition (equal); Methodology (equal); Resources (equal); Supervision (equal). **Zhe Cheng:** Conceptualization (equal); Data curation (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (equal); Visualization (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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